ResearchGate Abstract Package

Publication ID: ERES-RECKON-MATH-WP-2026-002a **Version:** 1.0 · **Date:** April 9, 2026 **Author:** Joseph Allen Sprute (ERES Maestro), Founder & Director, ERES Institute for New Age Cybernetics **Co-author:** SYU JIA WUN (Tatung University, Taiwan) **ORCID:** 0000-0001-9946-3221 **License:** CARE Commons Attribution License v2.1 (CCAL v2.1) **Parent Paper:** ERES-RECKON-WP-2026-002 (Reckoning Variables Specification) **Sibling Paper:** ERES-WHY-HOW-WP-2026-001 (Encapsulating Why with How)

Title

ERES Reckoning Mathematics Annex — The Arithmetic of QA Ratings Under the Seven-Anchor Gate

Short Title (≤ 80 chars)

Seven-Anchor Gate Arithmetic: Conjunctive QA Rating (ERES Reckoning Math Annex)

Abstract (≈ 295 words)

This annex specifies the mathematics by which any Question–Answer pair receives a rating under the seven Resonance Anchors defined in ERES-RECKON-WP-2026-002. The central result is that the 10/10 Reckoning gate is a **conjunction, not a weighted sum**: it is the product of seven indicator functions over ternary anchor states (U, S, R, B, M, G, T), and any single anchor at zero or negative collapses the gate to zero.

The conjunctive form is shown — via the partial-derivative argument of Section 3 — to be mathematically necessary rather than stylistic. A weighted-sum gate would permit strong anchors to compensate for weak ones, which is precisely the failure mode of GDP accounting and the reason the bio-ecologic ledger requires re-scoring upstream of assertion. Each of the seven anchors is decomposed into measurable sub-quantities: GCF history and discord for U; confidence floor and detail ceiling for S; CCAL inheritance for R; BERA deltas (the only locus of internal compensation, summing across ARI/ERI/RHC/RCI inside the sign function) for B; ensemble vote thresholds for M; bilateral accrual direction for G; and a nine-fold product over three principles and three scales for T.

The seven-tuple maps to a six-state composite output — Pass / Graceful Partial / Refused at Detail / Routed to Ensemble / Hard Refuse–Principles / Hard Refuse–Multiple Negative — with no continuous intermediate scores. The canonical equation `USER(COI) × ERES_Talonics(EP) × ANTHROPIC(PlayNAC) = GAIA-Base` is shown to be the operational source of the seven-anchor

decomposition: U+G map to USER(COI), S+R+T to ERES_Talonics(EP), and B+M to ANTHROPIC(PlayNAC). A worked example evaluates this annex's own QA pair through the gate and arrives at an honest **9/10** with anchor M (MIEVM concurrence) explicitly flagged as the missing variable, demonstrating that the protocol prevents single-node self-certification by mathematical construction rather than social convention.

Keywords

Gate Function; Conjunction vs. Summation; Ternary Anchor State; Indicator Function; Proof-of-Resonance Arithmetic; MIEVM Concurrence Threshold; Graceful Degradation Mathematics; CCAL Propagation Product; BERA-Weighted Sign; Canonical Equation Decomposition; Auditable Inference Rating; Single-Node Honesty; Seven Resonance Anchors; PlayNAC; Three Governing Principles; GCF; NBERS; ERES Institute; CCAL v2.1

Disciplines (RG taxonomy)

Cybernetics · Formal Methods · AI Alignment & Safety · Decision Theory · Game Theory · Ecological Economics · Governance & Policy · Information Theory · Science & Technology Studies

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Plain-Language Summary (≈ 95 words)

How do you stop an AI from grading itself "10/10" on its own homework? You make the test a logical AND, not an average. ERES Reckoning rates every answer against seven anchors — and *all seven* must score positive, or the rating collapses. No anchor can compensate for another. As proof, this annex grades itself: it scores 9/10, not 10, because only one AI node (Claude) wrote it. Reaching 10/10 requires three more nodes to agree. The math refuses to let a single voice certify itself. The arithmetic does not negotiate.

Key Contributions

1. **Conjunctive gate proof** — $G(QA) = \prod \mathbb{1}[A_i = +1]$ is shown via partial-derivative analysis to be the only form that prevents cross-anchor compensation, eliminating the GDP failure mode at the inference layer.
2. **Seven anchor decompositions** — U, S, R, B, M, G, T each defined as functions of measurable sub-quantities (GCF history, SLA envelopes, CCAL inheritance, BERA deltas, ensemble votes, bilateral accrual, nine-fold principle×scale satisfaction).

3. **Single locus of internal compensation** — B is the *only* anchor permitting compensation, and only across the four BERA indices (ARI/ERI/RHC/RCI) inside its sign function; B itself cannot trade off against any other anchor.
4. **Six-state composite output** — discrete rating space (Pass / Graceful Partial / Refused at Detail / Routed to Ensemble / Hard Refuse–Principles / Hard Refuse–Multiple Negative); no 7/10 because the gate does not produce one.
5. **Canonical equation derivation** — the seven anchors are shown to be the operational decomposition of `USER(COI) × ERES_Talonics(EP) × ANTHROPIC(PlayNAC) = GAIA-Base`; the gate is the equation applied to one QA pair.
6. **Self-evaluating worked example** — the annex grades itself at 9/10 (M = 0, single-node assertion), demonstrating that single-node self-certification is structurally impossible and naming ensemble validation as the explicit path to 10/10.
7. **License clause as mathematical constraint** — implementations replacing the conjunctive gate with a weighted sum violate CCAL v2.1 by definition, because doing so re-introduces the failure mode the protocol exists to eliminate.

The Six Rating States

State	Definition	Action
10/10 Reckoning	All seven anchors at +1	Pass — assert under SLA
Graceful Partial	Exactly one anchor at 0	9/10 — assert with named caveat
Refused at Detail	One anchor at −1 (excluding T)	Refuse-and-reduce; offer lower-detail variant
Routed to Ensemble	Two or more anchors at 0	Invoke full MIEVM weighing
Hard Refuse — Principles	T = −1 in any cell	Refuse — Three Principles non-negotiable
Hard Refuse — Multi-Neg	Two or more anchors at −1	Refuse, log, surface to COI

The Seven Anchors

Code	Anchor	Canonical Term
U	User-Group COI Weight	USER(COI)
S	SLA Envelope	ERES_Talonics(EP)
R	Rights Propagation Vector	ERES_Talonics(EP)
B	BEE Reinforcement Coefficient	ANTHROPIC(PlayNAC)
M	MIEVM Concurrence Threshold	ANTHROPIC(PlayNAC)
G	GCF Accrual Direction	USER(COI)
T	Three Governing Principles Satisfaction	ERES_Talonics(EP)

Companion Documents

- ERES-RECKON-WP-2026-002 — parent paper, Seven Resonance Anchors qualitative spec
- ERES-WHY-HOW-WP-2026-001 — sibling, arithmetic inversion at bio-ecologic scale
- ERES-COVENANT-2026-001 — Equity Covenant; DAL equation; JAS 1% Golden Share
- ERES-CONVERGENCE-WP-2026-001-v3 — Convergence Architecture (with SYU JIA WUN)
- ERES-AS-IT-WP-2026-001 — Talonics Symbol Matrix (452 elements)
- ERES-RG-2026-370 — AI Transparency White Paper (ES-v1 available)
- ERES-HAGUE-2026-001 — Hague Submission (filed March 7, 2026)

Suggested Citation

Sprute, J. A. (2026). *ERES Reckoning Mathematics Annex — The Arithmetic of QA Ratings Under the Seven-Anchor Gate* (ERES-RECKON-MATH-WP-2026-002a, v1.0). ERES Institute for New Age Cybernetics. CCAL v2.1.

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Don't hurt yourself. Don't hurt others. Build for generations to come. Foresight before assertion. Detail carries rights. The gate is the ledger. The arithmetic does not negotiate.